HARDWARE PROJECT REPORT

# Wavez

## A Campus FM Broadcasting Hardware System

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# 1. Abstract

Wavez is a dual-channel Campus Broadcasting hardware system built for Shanto-Mariam University of Creative Technology (SMUCT), centered on a Raspberry Pi 4 B as the broadcasting core and an Si4713 FM transmitter module as the RF output stage. The physical signal chain converts a digital media library into a clean, regulation-conscious analog FM broadcast covering a 400–700m radius, while the same core simultaneously outputs an internet-relayed stream for off-campus listeners. The hardware was deliberately chosen and assembled to avoid the RF integrity problems that plague DIY GPIO-based FM transmission approaches.

# 2. Problem Statement

SMUCT lacked any dedicated broadcasting hardware or physical transmission infrastructure for emergency notices, administrative alerts, and live announcements. Two hardware-level constraints shaped the design:

- **RF Interference Risk:** Transmitting directly from a Raspberry Pi's GPIO (e.g., PiFM-style square-wave modulation) produces uncontrolled harmonic emissions across a wide spectrum band, rather than a clean signal confined to the target frequency.
- **Fragmented Network Wiring:** With no unified campus intranet — different floors on different networks, SSIDs, and IP ranges via a third-party ISP — the hardware could not rely on automatic network-based switching between local and public output paths.

# 3. Objectives

- **Reliable Physical Transmission:** Build an RF signal chain that outputs a clean, regulation-conscious FM signal without harmonic bleed into adjacent bands.
- **Dual Output Hardware Path:** Have the same broadcasting core simultaneously drive a local FM transmitter and a public internet stream endpoint.
- **Field-Ready Coverage:** Achieve consistent 400–700m FM reception radius across campus grounds using compact, low-power hardware.

# 4. Hardware Architecture & Signal Chain

The Wavez hardware stack separates the digital processing core from the RF output stage, so the transmission hardware never has to touch unfiltered GPIO signal generation.

| Layer | Component / Hardware | Functionality |
|---|---|---|
| Broadcasting Core | Raspberry Pi 4 B | Central processing unit; runs the media/streaming software and drives the I2C bus to the transmitter. |
| FM Transmission | Si4713 Module | Onboard PLL and Low-Pass Filter shape a clean analog FM carrier over 400–700m. |
| Antenna Stage | Tuned Antenna & Cabling | Radiates the filtered FM signal; matched impedance avoids reflection losses. |

| Layer | Component / Hardware | Functionality |
|---|---|---|
| Power & Enclosure | Regulated Power Supply, Enclosure | Ensures stable voltage to the Pi and Si4713, and physical protection for continuous operation. |

### Signal Flow

Media Library on the Raspberry Pi 4B feeds the processing core, which drives the Si4713 module over I2C. The Si4713's internal PLL locks the carrier frequency while its low-pass filter strips harmonic content before the signal reaches the antenna stage, radiating a clean FM wave across the 400–700m target radius. In parallel, the same core pushes a digital stream out over the network for off-campus reception — two outputs, one controlled hardware core.

## 5. Component Specifications

| Component | Key Specification | Role in System |
|---|---|---|
| Raspberry Pi 4 B | Quad-core, 4GB RAM, I2C/GPIO interface | Runs broadcasting software; controls the Si4713 over I2C. |
| Si4713 Module | Integrated PLL, RDS support, low-pass output filter | Converts digital audio into a spectrum-clean analog FM carrier. |
| Antenna | Tuned length for FM band, coax-fed | Radiates the filtered signal; proper matching minimizes standing-wave losses. |
| MicroSD Storage | 32–64GB, Class 10 or better | Holds OS, media library, and buffered audio. |
| Power Supply | 5V/3A regulated USB-C | Prevents brownouts that could destabilize the PLL lock during transmission. |

## 6. Key Hardware-Enabled Features

### Clean, Filtered FM Output

The Si4713's onboard PLL and low-pass filter are the core hardware differentiator — they keep the transmitted signal confined to its intended frequency instead of leaking harmonics into adjacent bands, which is the main failure mode of unfiltered GPIO-based transmission approaches.

### Simultaneous Dual Output

A single Pi-driven core physically supports both the I2C-connected FM transmitter and a networked stream output at the same time, without needing separate broadcasting hardware for each path.

### Consistent Coverage Radius

Field testing confirmed a stable 400–700m reception radius, governed by the antenna's tuning and the Si4713's regulated output power rather than uncontrolled harmonic propagation.

## 7. Hardware Implementation & Assembly

The Raspberry Pi 4 B was set up as the physical hub, wired to the Si4713 module over its I2C bus for digital control of the FM carrier. Antenna length and placement were tuned specifically for the FM broadcast band, and cabling was chosen to minimize signal loss between the module and the radiating element. Choosing the Si4713 over direct GPIO-based transmission (e.g., PiFM) was a deliberate hardware decision to avoid harmonic interference — its onboard PLL and low-pass filtering keep the output confined to a clean signal. Ahead of deployment, the team reviewed BTRC guidance on low-power FM transmission to keep the broadcast hardware within commonly permitted limits for short-range transmitters of this kind. Power delivery was regulated to prevent voltage fluctuation from disturbing the PLL's frequency lock during continuous operation.

## 8. Testing & Field Results

The assembled hardware was field-tested directly on campus grounds. FM reception was confirmed within the targeted 400–700m radius using standard consumer FM receivers, with no observed interference artifacts outside the intended band. Power stability tests confirmed the regulated supply held the Si4713's PLL lock steady during extended continuous transmission, and the antenna/cabling assembly showed no measurable degradation over the testing period.

## 9. Project Timeline

| Phase | Weeks | Hardware Activities Completed |
|---|---|---|
| Planning & Procurement | 1–2 | Finalized requirements; procured the Pi, Si4713 module, antenna, and cabling. |
| Broadcasting Core Setup | 3–4 | Wired the Si4713 over I2C to the Pi; ran initial FM carrier tests. |
| Antenna & RF Tuning | 5–6 | Tuned antenna length and cabling; verified filter performance. |
| Power & Enclosure Build | 6–7 | Installed regulated power supply and physical enclosure for continuous operation. |
| Integration & Testing | 8 | End-to-end RF testing, signal cleanliness checks, stability testing. |
| Field Validation & Demo | 9–10 | On-campus FM range testing and final hardware demonstration. |

## 10. Challenges & Limitations

- **Avoiding Harmonic Interference:** Direct GPIO-based FM generation was ruled out early due to uncontrolled harmonic output; the Si4713's PLL and filtering hardware were required to keep the signal clean, adding component cost and I2C wiring complexity compared to a bare-wire approach.
- **No Automatic Local/Remote Switching:** Because there is no unified campus intranet, the hardware could not physically detect whether a listener's device was on-campus or off-campus, ruling out any network-hardware-based automatic switching between FM and internet output.
- **Power Stability:** Maintaining a steady PLL lock on the Si4713 required a properly regulated power supply, since voltage dips under load were found to destabilize the carrier frequency during testing.

## 11. Expected Impact & Future Scope

## Expected Impact

The Wavez hardware system gives SMUCT a physical, always-available broadcasting channel for Emergency and Academic notices, independent of the campus's fragmented network infrastructure.

## Future Scope

Planned hardware extensions include expanding the physical FM antenna infrastructure for deeper indoor penetration across all university floors, and evaluating higher-gain antenna configurations to extend the coverage radius without exceeding permitted transmission limits.